

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Classwork 4
  - Due **tomorrow** 10/11 @5 pm
- Midterm Question Contribution (Bonus Points!)
  - Google Form: https://forms.gle/PQWnzJ3EphU2WyPw8
  - Due **today** 10/10 @5 pm
- Homework 2 will be available today
  - Due Friday 10/25 @5 pm

This Lecture

- Approximate Inference in BN
  - Sampling in BN
  - Prior Sampling
  - Gibbs Sampling

Inference in BN

- Inference Goal of BN
  - $P(Q|e)$ given a BN (structure and semantics)
- Approaches
  - Exact Inference
  - Enumeration Inference
- Limitation
  - Computationally costly, why?
  - E.g., $P(Q, e) = \sum_y P(Q, e, y)$ -- a lot of hidden variables?

Precision vs. Efficiency

- Trade-off between precision and efficiency
- Simulation
  - A relatively small synthetic dataset to approximate the query
  - Sample vs. population
  - The distribution of the synthetic dataset is **consistent** with the true probability distribution.

Simulated Sampling

- Drawing Colored Marbles
  - A bag of marbles with colors C={R, G, B}
  - How to estimate the distribution of marbles in different colors, or P(C)?
    - Randomly sampling marbles from the bag
    - Sample size N
      - Higher N -> higher accuracy of the estimated P(C)
      - Vice versa

What is the limitation?

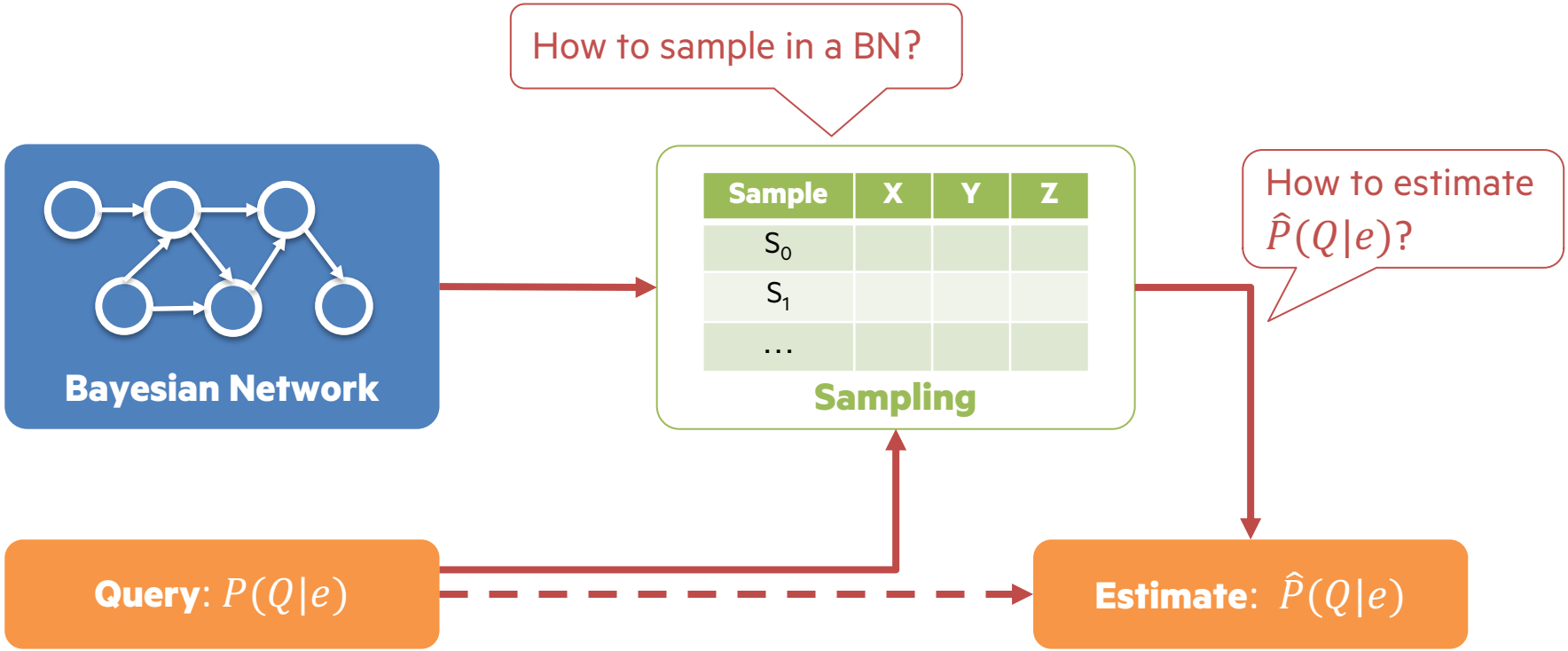
Law of Large Numbers

- As the sample size -> infinity, the likelihood of some event converges to its true likelihood. Namely, for the number N_s of samples of some event $N_s(x_1, x_2, \dots, x_i)$ (with total number of samples N), we have:

$$\lim_{N \to \infty} \frac{N_s(x_1, x_2, \dots, x_i)}{N} = P(x_1, x_2, \dots, x_i)$$

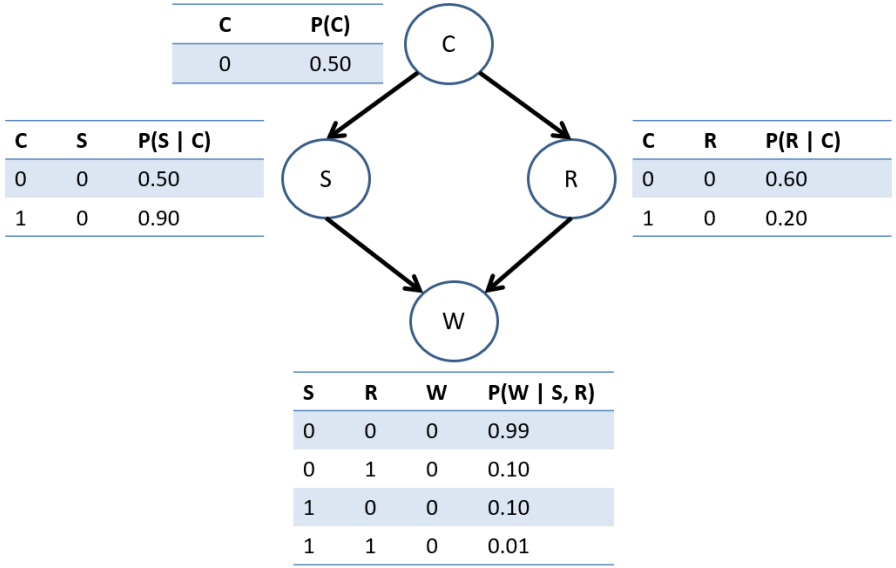
$$P(C) \approx \frac{N_s(C)}{N}$$

Sampling in a BN



Example

- Given a BN
  - C: cloudy
  - S: there are sprinklers on
  - R: raining
  - W: sidewalk is wet
- Estimate
  - $\hat{P}(W = 1) =$ 4/5
  - $\hat{P}(C = 1|W = 1) =$ 3/4



C	P(C)
0	0.50

C	S	P(S | C)
0	0	0.50
1	0	0.90

C	R	P(R | C)
0	0	0.60
1	0	0.20

S	R	W	P(W | S, R)
0	0	0	0.99
0	1	0	0.10
1	0	0	0.10
1	1	0	0.01

Sample	C	S	R	W
0	1	0	1	1
1	1	1	1	1
2	0	1	1	0
3	1	0	1	1
4	0	0	0	1

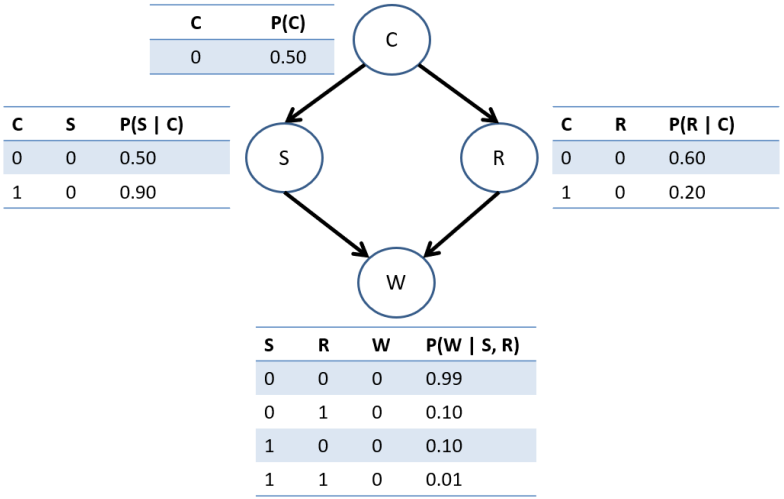
How to efficiently generate samples especially when P(e) is low?

Prior Sampling

- Given the BN structure and semantics, how to generate samples from the BN?
  - CPT: P(v|parents(v))
    - Sample **parents** first, then **child**
  - Proceduralize via **Topological Sort:**
    - an ordering of nodes in which a **node never appears before its parents** already have in the sequence.

Topological Ordering

- What are the possible topological sorts?
  - C, S, R, W
  - C, R, S, W



Order i	Given	Variable V_i	$P(V_i=0 \mid pa(V_i))$	Random Sample of V_i
0	{}	C	0.5	C=0
1	{C=0}	S	0.5	S=1
2	{C=0, S=1}	R	0.6	R=0
3	{C=0, S=1, R=0}	W	0.1	R=0

Repeat N times

Rejection Sampling

- What if the sample is **inconsistent** with the query's evidence?
  - Query: P(C|S=0)
  - Sample: {C=0, S=1, R=0, W=1}
- Rejection Sampling
  - Any sample inconsistent with evidence is immediately ignored / thrown out.

Gibbs Sampling

- What if the query's evidence is **rare**?
  - Query: P(C|S=0) but P(S=0)=0.000001
  - Solution: fix the evidence from the start
  - Sampling from some variable V given *all other* variables in the BN is much easier than sampling from the JPT itself.

Example

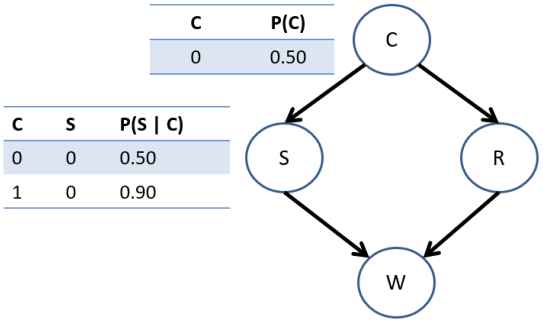
$$P(S|C,R,W) = \frac{P(S,C,R,W)}{P(C,R,W)}$$

$$= \frac{P(S,C,R,W)}{\sum_s P(S=s,C,R,W)}$$

$$= \frac{P(C)P(S|C)P(R|C)P(W|S,R)}{\sum_s P(C)P(s|C)P(R|C)P(W|s,R)}$$

$$= \frac{\cancel{P(C)}P(S|C)\cancel{P(R|C)}P(W|S,R)}{\cancel{P(C)}\cancel{P(R|C)}\sum_s P(s|C)P(W|s,R)}$$

$$= \frac{P(S|C)P(W|S,R)}{\sum_s P(s|C)P(W|s,R)}$$



C	P(C)
0	0.50

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C	R	P(R | C)
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0	0	0	0.99
0	1	0	0.10
1	0	0	0.10
1	1	0	0.01

Idea: fix the evidence, draw samples from a distribution where all other variables are given except for the ones are sampled.

Gibbs Sampling

- Initialize all variables to some random assignment, except for the evidence e, which remains fixed to its value.
- Sample a single variable at a time conditioned on the rest (again keeping e fixed).
- Repeat Step 2 many times.



Next…

MIDTERM Q&A

Feedback

- https://www.surveymonkey.com/r/RQKXLCN
- Brightspace -> Week 7

